

User Guide for DEV-CaMP Website (Updated June 3, 2025).

The purpose of **DEV-CaMP** is to provide researchers and practitioners with profiles of individual critical mineral mining projects located within the United States that will support the nascent critical materials industry in the United States. Although other databases provide a wealth of project-specific information (including geological and commercial variables), **DEV-CaMP** is uniquely rich on information about the federal financial support of specific projects, regulatory challenges, project-specific litigation, community and cultural concerns, public arguments made in favor of and in opposition to specific projects, and organizations that are active in public debates about specific projects.

For each Project entry, the various descriptive elements or variables are presented in a suite of thirteen categories or fields. Those fields are: Project Description, Location, Developer Description, Development Plans, Financial Support, Land Ownership, Statements of Support from Public Officials, Statements of Concern from Public Officials, Litigation, Concerned Interest Groups, Reasons for Concerns, Supportive Interest Groups, and Reasons for Support. Each field contains numerous subfields that are itemized below.

Each Project entry within the database contains relevant “Project Notes” about specific variables or subfields for that particular Project. The entries in the Notes are categorized in the same manner as the subfields appear in the database; each is grouped within and ordered in one of the thirteen fields.

This User Guide provides definitions of the terms used in the database and operational suggestions for users of **DEV-CaMP**. Since mining and processing are dynamic industrial sectors, **DEV-CaMP** will be updated periodically to reflect changes that occurred with the Projects listed and additions of new Projects. This version of **DEV-CaMP** covers data through January 2025. If users uncover errors or misleading or missing information in **DEV-CaMP**, they are encouraged to email grahamjd@iu.edu with correction requests.

DEV-CaMP covers post-2010 Projects (abandoned, active, or prospective) located in the United States that plan to mine/process one of seven critical materials that are used in batteries, motors, and other electronic/electrical systems in electric vehicles, as well as other applications in the energy transition. These materials are: cobalt, copper, graphite, lithium, manganese, neodymium, and nickel. Of these seven minerals, all but copper have been deemed as “critical minerals” by the U. S. Geological Survey. Copper is included in the U. S. Department of Energy’s listing of “critical materials”, therefore all are described in this database as critical materials. In the future, the database may be expanded to include additional materials and additional projects in North America.

DEV-CaMP fields are being made publicly available as the IU research team publishes papers using these data. As of June 3, 2025, over half of the data are publicly available. If users are interested in accessing data in selected fields not yet released, they should send a request to grahamjd@iu.edu.

There are a total of 13 fields in **DEV-CaMP**, with each field including up to 17 subfields. The fields and subfields are defined below.

Fields in DEV-CaMP:

1. Project Description (7 subfields):

Primary Critical Material: The “primary” material is that commodity that has been highlighted by the developer and coincides with one of the seven minerals that are being included in this database. The primary material is not necessarily the most commercially important material for a given Project.

Project Name: The name of the Project shown is the term that the developer has assigned to an effort to explore, develop, or produce a deposit. If the name of a project was changed recently, it is documented in a note on the project’s page.

Mining Method: The mining method is the technique used to extract the ore, or fluid, that contains the material. The methods are categorized as: surface, underground, surface and underground, recovery from waste, evaporation of brines, direct lithium extraction, and undetermined. Notes are used to elaborate on projects that are using a unique extraction method. For projects in the exploration stage, the mining method is often not yet determined.

Co-Producing Materials: The co-produced materials are a list of the additional materials (beyond the primary critical material) that the developer is mining or expects to obtain from the site. The co-produced materials may or may not be critical materials as defined by the US Department of Energy.

Existing Mine: This subfield documents whether the site where the Project is located was previously mined (e.g., abandoned), is currently being mined, or is a greenfield site. Notes are sometimes used to provide non-standardized information about the mining and/or industrial history of the site.

Status of Project: The classification for Project status are: exploration, development, active production (with or without an expansion plan), and delayed indefinitely. As active Projects reach the closure stage, a category for closure/reclamation may be added. Projects that are cancelled are not distinguished from those that are delayed indefinitely. A delay is deemed to be indefinite if the developer is no longer working on the site or has entered the development stage but does not have a transparent schedule to begin production, or if the developer is in the exploration stage and has encountered a formidable commercial and/or regulatory barrier that leads to protracted delay. Notes are used to document cases where a developer has announced an indefinite delay.

Regulatory Challenges: A regulatory challenge occurs if a project has been significantly delayed or altered due to a regulatory complication (e.g., concerns of the local community, land ownership or water disputes, permitting difficulties and litigation). Some regulatory challenges are current while others may have occurred years ago (but after 2010) and are not a current issue.

2. Location (7 subfields): This field provides various information on where the project is located.

State: The state refers to one of the 50 US states where the project is located. The database does not yet include any projects that cross state boundaries. The processing facility may not necessarily be co-located with the mining site. For information on the location of the processing facility, consult the fields under “Commercial Plans”.

County: The county is a governance unit within a state that provides certain kinds of services. Most states are organized by counties (Alaska, which does not have counties, is an exception).

Nearest Population Center: The nearest population center is the closest town, village or city to the project. Additional location information, including distances to major population centers, is located in the notes section.

Population Size: The size of the nearest population center's population is typically drawn from the 2020 Census in the US.

Additional Location Information: These notes provide non-standardized information about the location of each project such as region of state (north, south, east, west), proximity to major population centers, proximity to bordering states, proximity to highways, proximity to nearby mines (active, planned or abandoned), and proximity to parks, waterbodies and/or forests.

Mining District: A mining district is a special-purpose administrative subdivision used in North America. They were initially developed as a means of self-governance for 19th-century prospectors and were later defined by statute. Some projects are not located in a mining district. For areas where minerals are known to occur and have been produced, but no formal mining districts have been established, the term "mineral occurrence" defines the geographic and geologic area where mining has occurred.

Mineral Occurrence: Mineral occurrence refers to a geological trend and associated geographical area that includes deposits of a given commodity.

Developer Description (6 subfields):

Lead Developer: The company serving as the lead developer heads the design and execution of the project, typically with a high rate of ownership. It may be a small start-up company, a large mining company or its subsidiary, or an aggregation of several companies. If the project is a joint venture of two or more companies, the joint venture is described in a Project Note and the "lead developer" is typically that has the dominant financial interest in the Project.

Location of the Lead Developer's Headquarters: This subfield provides the country where the lead developer's corporate headquarters is located.

Lead Developer's Website: A link is provided either to the lead developer's corporate website or to a website on the project prepared by the lead developer.

Parent Company: In cases where the lead developer is a subsidiary of a parent company, this subfield provides the name of the parent company.

Location of the Parent Company's Headquarters: If applicable, this note provides the country where the parent company's corporate headquarters is located.

Was There a Change in Ownership during the Project?: This subfield indicates whether there has been a recent change in project ownership. Notes are used to describe recent changes.

3. Development Plans (6 subfields):

Planned Production Capacity (Thousands of Tons/Year): This subfield provides the amount of material to be mined (thousands of tons per year), as specified in a Preliminary Economic Assessment or another technical report sponsored or prepared by the developer. Such information is often not available for projects in the exploratory stage. For active mines, production volume for a recent year is provided; notes describe any expansion plans.

Planned Lifespan of Mine (Years): This subfield describes the number of years that the mine will likely be active, given the size of the deposit and the planned annual production volume. This information is typically taken from a Preliminary Economic Assessment (PEA) or other technical document sponsored or prepared by the developer. Such information is often not available for projects in the exploratory stage.

Processing Plan in the US: This subfield indicates whether the developer is processing the material in the United States or has a public plan to process in the United States.

Processing Plan Elsewhere in North America – not yet published: This subfield indicates whether the developer is processing the material elsewhere in North America (e.g., Canada) or has a public plan to do so.

Processing Done by the Same Lead Developer/Parent Company: This subfield indicates whether the processing is, or is going to be, performed by the same company (lead developer or parent) that is doing the mining.

Total Capital Costs (Millions of Dollars): For projects in the development stage, this subfield provides an estimate of the capital costs for the planned project (mining and processing), typically as reported in the Preliminary Economic Assessment or a related technical document sponsored by or prepared by the developer. Projects at the exploration stage rarely have capital cost estimates for the full project. Capital costs are not relevant to active mines unless there are expansion plans, in which case any public estimates of capital costs are reported.

4. Financial Support (7 subfields):

Number of Federal Awards: The number of federal awards (grants or loans) received by the developer for the project.

Federal Agency: DOD, DOE and/or EXIM.

Year Granted: The year the award was announced, which is not always the year the award took effect.

Grant/Loan: A grant is a direct subsidy while a loan has a payback requirement influenced by the interest rate and term of the loan.

Amount: The overall size of the grant or loan, usually the loan is technically a loan guarantee.

Purpose of Support: This note indicates what the financial support is for (e.g., conducting a feasibility analysis, building a processing facility or purchasing mining equipment).

Public Supply Agreement with an Automaker or Battery/Component Producer: This subfield (yes/no) indicates whether the developer has publicized any collaborative arrangements (e.g., supply commitments or joint ventures) with an automaker or battery/component producer that have revenue ramifications for the project. Notes may describe some of the notable arrangements.

5. Land Ownership (3 subfields):

Type of Land: This subfield indicates whether the land on which the project is being developed is public, private, or a combination of public and private land.

Public Land Ownership: This subfield, where applicable, indicates whether the public land is owned by the federal government, the state government, or a combination. For federal land, the subfield indicates whether the land is overseen by the US Bureau of Land Management (BLM) in the Department of Interior or the US Forest Service (USFS) in the Department of Agriculture.

Year of Land Acquisition: This subfield indicates the year the developer acquired the rights to the land.

6. Statements of Support from Public Officials (3 subfields – not yet published):

Local: This subfield indicates (yes/no) that we found that a local government official made a publicly available supportive statement about the project.

State: This subfield indicates (yes/no) that we found that a state official made a publicly available supportive statement about the project.

Federal: This statement indicates (yes/no) that we found that a federal official made a publicly available supportive statement about the project.

7. Statements of Concern from Public Officials (3 subfields – not yet published):

Local: This subfield indicates (yes/no) that we found that a local government official made a publicly available concerned statement about the project.

State: This subfield indicates (yes/no) that we found that a state government official made a publicly available concerned statement about the project.

Federal: This subfield indicates (yes/no) we found that a federal government official made a publicly available concerned statement about the project.

8. Litigation (6 subfields):

Evidence of Litigation: This subfield indicates (yes/no) whether we uncovered any evidence of post-2010 lawsuits related specifically to the project. Litigation does not

necessarily lead to a judicial decision since parties often settle; since settlements are often confidential, no data is provided on settlements.

Number of Cases: This subfield indicates the number of lawsuits we found related to the project, including both active and resolved cases.

State and/or Federal: This subfield indicates whether the litigation was in state courts, federal courts, or both.

Court Rulings Issued: This subfield indicates (yes/no) if court rulings post-2010 were issued concerning lawsuits related specifically to the project. A layman's explanation of the ruling is found in the notes section.

Level of Court: District, Appeals and/or Supreme Court.

Link to Judicial Decision: If applicable, this note contains a link to the most recent judicial decision in the notes section.

- 9. Concerned Interest Groups (8 subfields – not yet published):** This field categorizes the private individuals/groups that raised concerns about the project. Evidence for a particular group's interest in a project is denoted by a check, further details are contained in the notes section. Groups include the following subfields:

Community Residents/ Groups

Indigenous Groups

Environmental Groups

Scale of Environmental Group(s): Local/regional, national, or both.

Commercial Groups: E.g., individual businesses and business associations.

Recreational Groups: E.g., associations of hikers, hunters, fishermen, etc.

Other

Evidence of Organized Protest(s): This field indicates whether we found any evidence of organized public protests of the project.

- 10. Reasons for Concern (17 subfields – not yet published):** This field indicates the types of reasons for concern about the project that we uncovered. These concerns were not reviewed for validity, and are not comprehensive as evidence was sourced from local news media in the project area.

Land Use (Built Environment)

Impacts to Natural Environment

Site of Historical, Cultural, Spiritual Significance

Traffic & Noise

Air Quality

Water Supply

Water Quality

Impact on Species

Classification of Species Impacted: Threatened and/or endangered, where applicable.

Recreation

Agriculture

Subsistence Hunting/Fishing

Historical Mining or Reclamation Issues

Reputation of the Developer/Parent Company – not yet published

Inadequate Community Engagement: E.g., by the developer, local government, state government or federal agencies.

Waste Management

Other

11. Supportive Interest Groups (3 subfields – not yet published): This field indicates which groups made public statements supportive of the project. We do not include the developer here because we assume they are always a supportive group.

Community Residents/Groups

Commercial Groups: E.g., individual businesses and business associations.

Other

12. Reasons for Support (10 subfields – not yet published): This field indicates the types of reasons for support found in local news articles. As with reasons for concern, we have not determined the validity of the reasons. They are extracted from local news articles and developer communications.

Economic Growth

Tax Revenue

Employment

Community Development

Replace Lost Industries

Historical Mining Experience: I.e., a favorable history with mining in the area.

Geopolitical/Security: E.g., need for diminished dependence on foreign sources of material such as China or countries with unstable governance regimes or lack of respect for human rights.

Clean Energy Transition: E.g., need for materials to support energy technologies that will mitigate climate change and/or enhance regional or local air quality.

Cleaner than Other Sources: E.g., mining and processing at this site is cleaner or safer than it would be at other sites in the US or abroad.

Other